

Experiment 2

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Title: Implementation of SELECT Queries with Filtering, Grouping and Sorting in PostgreSQL

1. Aim:

- a. To implement and analyze SQL SELECT queries using filtering, sorting, grouping, and aggregation concepts in PostgreSQL for efficient data retrieval and analytical reporting.

2. Software Requirements:

- a. PostgreSQL (Database Server)
- b. pgAdmin
- c. OS: Windows

3. Objective:

- a. To retrieve specific data using filtering conditions
- b. To sort query results using single and multiple attributes
- c. To perform aggregation using grouping techniques
- d. To apply conditions on aggregated data
- e. To understand real-world analytical queries

4. Procedure of the Practical:

- a. Start the system.
- b. Open pgAdmin.
- c. Create and select the database in which you want to perform the experiment.
- d. Establish connection to the database using Alt+Shift+Q.
- e. Run the following queries given in Experiment Steps (5).

5. Experiment Steps:

a. Create table queries:

- Create table customer_orders (order_id int primary key, customer_name varchar(30) not null, product varchar(50) not null, quantity int not null check (quantity > 0), price decimal(8, 2) not null check (price >= 0), order_date date not null);

b. Insert queries:

- Insert into customer_orders values (1023, 'amit sharma', 'laptop', 1, 55000.00, '2024-01-05');

	order_id [PK] integer	customer_name character varying (30)	product character varying (50)	quantity integer	price numeric (8,2)	order_date date
1	1023	Amit Sharma	Laptop	1	55000.00	2024-01-05
2	2045	Neha Verma	Mouse	2	750.00	2024-01-06
3	3098	Ravi Kumar	Keyboard	1	1500.00	2024-01-07
4	4121	Priya Singh	Laptop	2	52000.00	2024-01-08
5	5876	Ankit Patel	Monitor	1	12000.00	2024-01-09
6	6234	Sneha Iyer	Mouse	3	700.00	2024-01-10
7	7459	Rahul Mehta	Keyboard	2	1400.00	2024-01-11
8	8562	Pooja Nair	Monitor	2	11500.00	2024-01-12
9	9187	Vikas Gupta	Laptop	1	58000.00	2024-01-13
10	9904	Kiran Rao	Mouse	1	800.00	2024-01-14

c. Filtering Data using conditions

- Select * from customer_orders where price > 45000;

	order_id [PK] integer	customer_name character varying (30)	product character varying (50)	quantity integer	price numeric (8,2)	order_date date
1	1023	Amit Sharma	Laptop	1	55000.00	2024-01-05
2	4121	Priya Singh	Laptop	2	52000.00	2024-01-08
3	9187	Vikas Gupta	Laptop	1	58000.00	2024-01-13

d. Sorting Query Results:

- Sorting in descending order:

Select order_id, customer_name as name, price as shopping_value from customer_orders order by shopping_value desc;

	order_id [PK] integer	name character varying (30)	shopping_value numeric (8,2)
1	9187	Vikas Gupta	58000.00
2	1023	Amit Sharma	55000.00
3	4121	Priya Singh	52000.00
4	2678	Meena Joshi	12500.00
5	6802	Manoj Kulkarni	12000.00
6	5876	Ankit Patel	12000.00
7	5791	Divya Kapoor	12000.00
8	8562	Pooja Nair	11500.00

ii. Priority based sorting:

Select order_id, customer_name as name, price as shopping_value from customer_orders order by shopping_value desc, order_id;

	order_id [PK] integer	name character varying (30)	shopping_value numeric (8,2)
1	9187	Vikas Gupta	58000.00
2	1023	Amit Sharma	55000.00
3	4121	Priya Singh	52000.00
4	2678	Meena Joshi	12500.00
5	5791	Divya Kapoor	12000.00
6	5876	Ankit Patel	12000.00
7	6802	Manoj Kulkarni	12000.00
8	8562	Pooja Nair	11500.00

e. Grouping data for aggregation:

i. Select product, count(product) as quantity_sold, sum(quantity * price) as total_sales from customer_orders group by product;

	product character varying (50)	quantity_sold bigint	total_sales numeric
1	Mouse	9	6650.00
2	Keyboard	9	12850.00
3	Monitor	7	83500.00
4	Laptop	4	217000.00

f. Applying conditions on aggregate data:

i. Select product, sum(quantity) as quantity_sold, sum(quantity * price) as total_sales from customer_orders group by product having sum(quantity) >= 5;

	product character varying (50)	quantity_sold bigint	total_sales numeric
1	Mouse	9	6650.00
2	Keyboard	9	12850.00
3	Monitor	7	83500.00

g. Filtering vs Aggregation Conditions

i. Filtering: Select product, sum(quantity * price) as total_sales from customer_orders where quantity >= 2 group by product;

	product character varying (50)	total_sales numeric
1	Keyboard	9850.00
2	Monitor	47000.00
3	Mouse	5100.00
4	Laptop	104000.00

- ii. Aggregation Conditions: Select product, sum(quantity * price) as total_sales from customer_orders group by product having sum(quantity * price) > 50000;

	product character varying (50) 🔒	total_sales numeric 🔒
1	Mouse	6650.00
2	Keyboard	12850.00
3	Monitor	83500.00
4	Laptop	217000.00

6. Input/Output Details:

- The input for the experiment is the SQL queries mentioned in Experiment Steps (5).
- The output for each query is attached after the query itself.

7. Learning Outcome

- Understand how data can be filtered to retrieve only relevant records from a database.
- Learn how sorting improves readability and usefulness of query results in reports.
- Gained the ability to group data for analytical purposes.
- Differentiate between row-level conditions and group-level conditions.
- Writing analytical SQL queries used in real-world scenarios.
- Better prepared to answer SQL-based questions related to filtering, grouping, and aggregation.